

Data Preprocessing Report

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Heart Disease Prediction Dataset

1. Introduction

This project focused on preparing the Heart Disease Prediction Dataset for machine learning. The dataset contains **918 patient records and 12 original variables**, including demographic information, clinical measurements, ECG results, exercise-related indicators, and the binary target variable **HeartDisease**. The objective of preprocessing was to transform the raw data into a clean, consistent, numerical dataset while preserving clinically meaningful information and avoiding unnecessary removal of observations.

The preprocessing workflow included data inspection, data cleaning, exploratory visualisation, feature engineering, categorical encoding, outlier assessment, feature scaling, correlation analysis, and feature importance analysis.

2. Data Inspection and Initial Findings

Initial inspection confirmed that the dataset contained **918 observations and 12 variables**. There were no conventional missing values recorded in the dataset, and the variables consisted of **six integer columns, one floating-point column, and five categorical/object columns**.

The numerical variables included **Age**, **RestingBP**, **Cholesterol**, **FastingBS**, **MaxHR**, **Oldpeak**, and the target **HeartDisease**. The categorical variables were **Sex**, **ChestPainType**, **RestingECG**, **ExerciseAngina**, and **ST_Slope**.

Descriptive statistics were examined to understand the distributions and identify potentially problematic values. **Age** ranged from 28 to 77 years, while **RestingBP** ranged from 0 to 200 and **Cholesterol** from 0 to 603. The zero values in **RestingBP** and **Cholesterol** required particular attention because they are not clinically plausible measurements.

The target variable was reasonably balanced, with **508 patients (55.3%) having heart disease and 410 (44.7%) without heart disease**. Therefore, severe class imbalance was not considered a major preprocessing concern.

3. Exploratory Visualisation

Several visualisations were used to understand the structure of the data before preprocessing.

Target Variable Distribution

The target distribution showed a relatively balanced classification problem, with 55.3% of observations belonging to `HeartDisease = 1` and 44.7% to `HeartDisease = 0`.

Categorical Feature Distributions

The categorical distributions showed that the dataset was predominantly male and that some categories were considerably more common than others. `ASY` was the dominant `ChestPainType`, while `Normal` was the most common `RestingECG` category. `FastingBS = 0` and `ExerciseAngina = N` were also more frequent than their alternatives.

These distributions were retained rather than artificially balanced because unequal category frequencies do not automatically indicate poor-quality data.

Categorical Features Against Heart Disease

Further analysis compared `ChestPainType`, `ExerciseAngina`, and `ST_Slope` against `HeartDisease`.

`ChestPainType` showed substantial differences in heart disease prevalence. The `ASY` category had a heart disease rate of approximately **79.0%**, compared with **13.9% for ATA**.

`ExerciseAngina` also showed a strong relationship with the target. Patients with exercise-induced angina had a heart disease rate of approximately **85.2%**, compared with **35.1%** among patients without exercise-induced angina.

`ST_Slope` showed another strong relationship. The heart disease rate was approximately **82.8% for Flat**, **77.8% for Down**, and only **19.7% for Up**.

These findings indicated that these categorical variables contained potentially useful predictive information and should therefore be retained.

Numerical Feature Analysis

Boxplots, distributions, and correlation analysis were used to examine the numerical variables.

`Oldpeak` showed a positive relationship with heart disease, while `MaxHR` showed a negative relationship. The approximate correlations with the target were **+0.40 for Oldpeak** and **-0.40 for**

MaxHR. Patients with heart disease generally showed higher and more widely distributed **Oldpeak** values, while patients without heart disease generally had higher **MaxHR** values.

A correlation heatmap was also used to examine relationships among the numerical variables and identify potential multicollinearity.

4. Data Cleaning

Invalid Values

Two clinically implausible zero-coded values were identified:

- **RestingBP = 0: 1 record**
- **Cholesterol = 0: 172 records (18.7%)**

These values were treated as missing rather than as genuine measurements. They were converted to **NaN** and replaced using the **median of the valid observations**.

Median imputation was selected because it is less affected by extreme values than mean imputation and preserves all 918 patient records.

Importantly, the records themselves were **not deleted**. Only the invalid values were corrected.

Duplicates and Data Types

Duplicate records were assessed and no records required removal. Data types were also validated to ensure that numerical variables were numeric and categorical variables were suitable for encoding.

5. Feature Engineering

Feature engineering was considered based on the available clinical variables.

No additional derived features were created because the existing variables already represented meaningful clinical measurements and categories. Creating artificial combinations without sufficient clinical justification could introduce unnecessary complexity.

No irrelevant features were identified for removal. All **11 original predictor variables** were retained because each provided potentially useful information for predicting heart disease.

The original column names were also retained because they were already clear, descriptive, and consistently formatted.

6. Categorical Encoding

Categorical variables were converted into numerical representations suitable for machine learning.

`Sex` and `ExerciseAngina` were converted using **binary encoding**, while `FastingBS` was already represented as a binary numerical variable.

`ChestPainType`, `RestingECG`, and `ST_Slope` were encoded using **One-Hot Encoding with `drop_first=True`**.

One-hot encoding was selected because these categories are nominal rather than naturally ordered. Using simple label encoding could incorrectly imply a ranking between categories. Dropping one reference category also avoids unnecessary duplicate information among dummy variables.

The encoding increased the dataset from 12 original columns to **16 final columns: 15 predictors and 1 target variable**.

7. Feature Scaling

The continuous numerical predictors were standardized using `StandardScaler`:

- `Age`
- `RestingBP`
- `Cholesterol`
- `MaxHR`
- `Oldpeak`

Before scaling, these variables had substantially different ranges and measurement units. After scaling, their means were approximately **0** and standard deviations approximately **1**.

For example, `Cholesterol` originally had a mean of approximately **243.20** after invalid-value correction, while `Age` had a mean of approximately **53.51**. Standardization placed these variables on a comparable numerical scale.

Binary and one-hot encoded variables were not scaled because their 0/1 representation is already appropriate for machine learning.

8. Outlier Detection and Treatment

Outliers were investigated using **boxplots and the IQR method**.

Several statistically extreme values were identified, including high `RestingBP`, high `Cholesterol`, low `MaxHR`, and negative or high `Oldpeak` values. However, statistical extremeness alone was not considered sufficient evidence that a value was erroneous.

The clinically invalid zero values in `RestingBP` and `Cholesterol` were therefore corrected, while other extreme observations were retained because they could represent genuine patient measurements.

This approach avoided automatically deleting potentially meaningful clinical observations.

9. Correlation and Multicollinearity Analysis

Correlation analysis was first performed on the numerical variables. `Oldpeak` and `MaxHR` showed the strongest relationships with `HeartDisease` among the continuous variables, with correlations of approximately **+0.40** and **-0.40**, respectively.

Correlation analysis was also performed after categorical encoding to assess the relationship between individual encoded categories and the target. `ST_Slope_Up` ($r \approx -0.62$) and `ST_Slope_Flat` ($r \approx +0.55$) showed the strongest individual correlations among the encoded categorical predictors.

A strong negative correlation of approximately **-0.87** was observed between `ST_Slope_Flat` and `ST_Slope_Up`. This was expected because both columns originate from the same

categorical variable and represent mutually exclusive categories. Therefore, it does not indicate problematic multicollinearity between independent clinical measurements.

Overall, no severe problematic multicollinearity requiring feature removal was identified.

10. Feature Importance Analysis

A preliminary Random Forest classifier was used as a complementary method for assessing feature importance.

The model achieved:

- **Training accuracy: 100%**
- **Testing accuracy: 87.5%**

The difference suggests some degree of overfitting; however, this model was used only as a preliminary feature-importance tool and not as the final predictive model.

The feature importance ranking was:

Feature	Importance
ST_Slope_Up	0.1604
ST_Slope_Flat	0.1220
MaxHR	0.1143
Oldpeak	0.1109
Cholesterol	0.0858
Age	0.0843
ExerciseAngina	0.0784
RestingBP	0.0685
ChestPainType_ATA	0.0395

Sex

0.0371

The results reinforce the earlier EDA findings, particularly the importance of **ST_Slope**, **MaxHR**, and **Oldpeak**. However, feature importance was not used alone to remove variables. Since all predictors contained potentially useful information and no feature demonstrated clear redundancy or invalidity, **no predictors were removed**.

11. Final Machine-Learning-Ready Dataset

The final preprocessing produced a dataset containing:

- **918 patient records**
- **16 columns**
- **15 predictors**
- **1 target variable**
- **0 missing values**
- **0 duplicate records**
- Categorical variables numerically encoded
- Continuous variables standardized
- Invalid zero-coded measurements corrected
- Clinically plausible extreme observations retained

The resulting dataset is therefore suitable for the **next stage of model development and evaluation**.

A future modelling pipeline should fit imputation and scaling parameters using the training data only before evaluating the final model, in order to prevent data leakage.

12. Overall Conclusion

The preprocessing stage successfully transformed the raw Heart Disease Prediction Dataset into a structured machine-learning-ready dataset. Invalid clinical placeholder values were corrected, categorical variables were appropriately encoded, numerical features were standardized, and potential outliers and correlations were assessed without unnecessarily removing valid observations.

EDA and feature-selection analyses consistently highlighted **ST_Slope, ExerciseAngina, Oldpeak, and MaxHR** as particularly informative predictors. No features were removed because there was insufficient evidence that any variable was irrelevant or genuinely redundant. The final dataset therefore provides an appropriate foundation for developing and evaluating a supervised heart disease classification model